UniMind ■ - Phase 2 Implementation Plan

Subsystem: Document Ingestion, PyMuPDF Parsing & Metadata Engine

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1. Core Architectural Modules & Planned Files

Module / Layer	Target File	Operational Responsibility
Dependencies	requirements.txt	Add python-multipart>=0.0.9 for FastAPI form-data handling.
File Security	backend/app/utils/file_validator.py	Magic bytes check (b'%PDF-'), MIME type, size limit, filename sanitization.
Text Cleaning	backend/app/rag/cleaner.py	Unicode NFKC normalization, hyphenation restoration across line breaks, whitespace cleanup.
PyMuPDF Extractor	backend/app/rag/parser.py	In-memory extraction, structural text blocks, scanned page / image detection, empty page checks.
Document Service	backend/app/services/document_service.py	Orchestrates upload workflow, raw storage (data/raw/), processed storage (data/processed/).
Schemas	backend/app/models/schemas.py	DTOs for upload requests, document metadata, page statistics, and error models.
API Endpoints	backend/app/routes/documents.py	POST /api/v1/documents/upload, GET /api/v1/documents/{id}, GET /api/v1/documents.
Test Suite	tests/test_pdf_parser.py tests/test_documents_api.py	Automated tests using strictly synthetic generated PDFs (standard, empty, corrupt, scanned).

2. Security, Scanned Documents & Edge Cases

Edge Case / Security Threat	Defensive Strategy & Implementation
File Type Spoofing	Validates first 5 bytes against PDF signature %PDF-. File extension alone is never trusted.
Path Traversal / Injection	Filename sanitized to alphanumeric characters; stored on disk using deterministic UUID4 keys.
Denial of Service (Oversized)	Enforces strict 25 MB file limit. Rejects excess payloads immediately with HTTP 413.
Encrypted / Locked PDFs	Checks doc.is_encrypted prior to extraction. Returns HTTP 422 with clear diagnostic message.
Scanned Image PDFs	Detects pages where extractable text < 50 chars but image count > 0. Flags document with scanned warning.
Empty Documents	Detects 0-length text; marks status as EMPTY to prevent polluting downstream vector indexes.

3. Synthetic Testing Strategy & Implementation Steps

Synthetic Document Protocol: To strictly adhere to university compliance constraints, tests will dynamically synthesize test PDFs (Standard Policy Mock, Blank Document, Corrupted Byte Stream, Image-only Mock). Real institutional documents are completely excluded.

Step-by-Step Implementation Sequence:

1. **Step 2.1:** Add python-multipart>=0.0.9 to requirements.txt & verify environment.
2. **Step 2.2:** Build file_validator.py (magic bytes, MIME check, filename sanitization).
3. **Step 2.3:** Build cleaner.py (NFKC normalization, hyphenation restoration, whitespace cleaning).
4. **Step 2.4:** Build parser.py using PyMuPDF (block extraction, page metadata, scanned detection).
5. **Step 2.5:** Build document_service.py (persistence to data/raw/ and data/processed/).
6. **Step 2.6:** Define Pydantic request/response schemas in schemas.py.
7. **Step 2.7:** Build documents.py router and mount under /api/v1/documents.
8. **Step 2.8:** Author synthetic test suites in tests/ and run verification.
9. **Step 2.9:** Generate Milestone PDF and Markdown completion reports and await approval.